User Manuel on the tool "PS-SAT: Predictive Scheduling & Satisfaction Analytics Tool"

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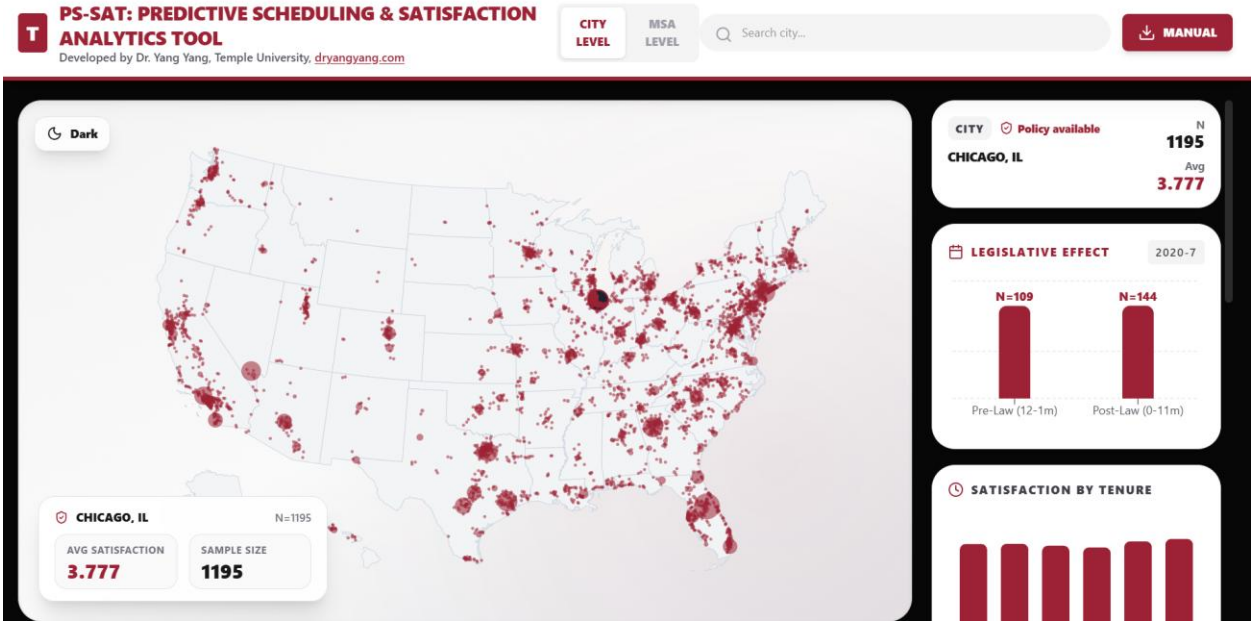
Based on: Yang, Y., Tan, K. & Liu, Y. (forthcoming). Balancing the Clock: How Predictive Scheduling Laws Influence Employee Satisfaction in the U.S. Tourism and Hospitality Industry. *Tourism Management*.



1. Introduction & Purpose

This web-based dashboard is designed to visualize tourism and hospitality employee satisfaction data across U.S. geographic units at both the City and Metropolitan Statistical Area (MSA) levels. The application provides interactive mapping, subgroup breakdowns, and (where applicable) legislative pre– and post–comparisons.

Key Features:

• **Dual-Level Geographic View:** Users can switch between City-Level (point markers) and MSA-Level (polygon map) visualization.

• **Interactive Search & Selection:** Users can search locations dynamically and select them either via the map or the dropdown search interface.

• **Subgroup Breakdown Analysis:** Satisfaction scores are displayed by:

- Job tenure
- Business model

- Skill level
- Front-of-house vs. back-of-house

• **Legislative Effect Comparison:** For locations with valid policy-period data, the dashboard displays Pre-Law vs. Post-Law satisfaction comparisons.

• **Visual Encoding of Sample Size:** City marker size scales with sample size (N), allowing immediate identification of data-rich locations.

2. Step-by-Step Usage Guide

Step 1: Select Geographic Level

At the top-left navigation bar, select:

• **CITY LEVEL**
Displays cities as scalable markers on a U.S. map.

• **MSA LEVEL**
Displays Metropolitan Statistical Areas as clickable polygons.

The selection determines how geographic units are rendered and interacted with.

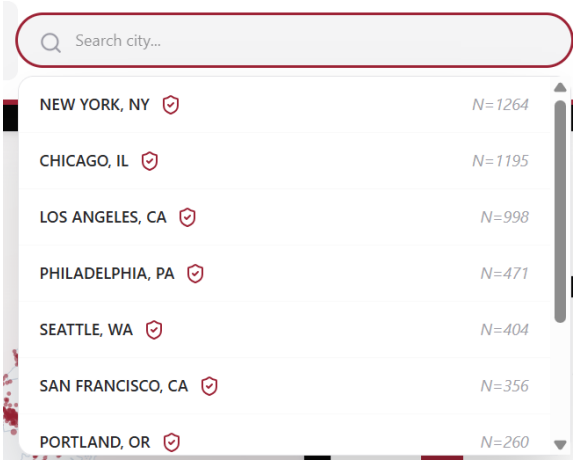
Step 2: Search or Select a Location

You may select a location in two ways:

Option A: Use the Search Bar

• Click inside the “Search database...” box.
• Begin typing a city or MSA name.
• A recommendation list appears dynamically.
• Click any result to select it.

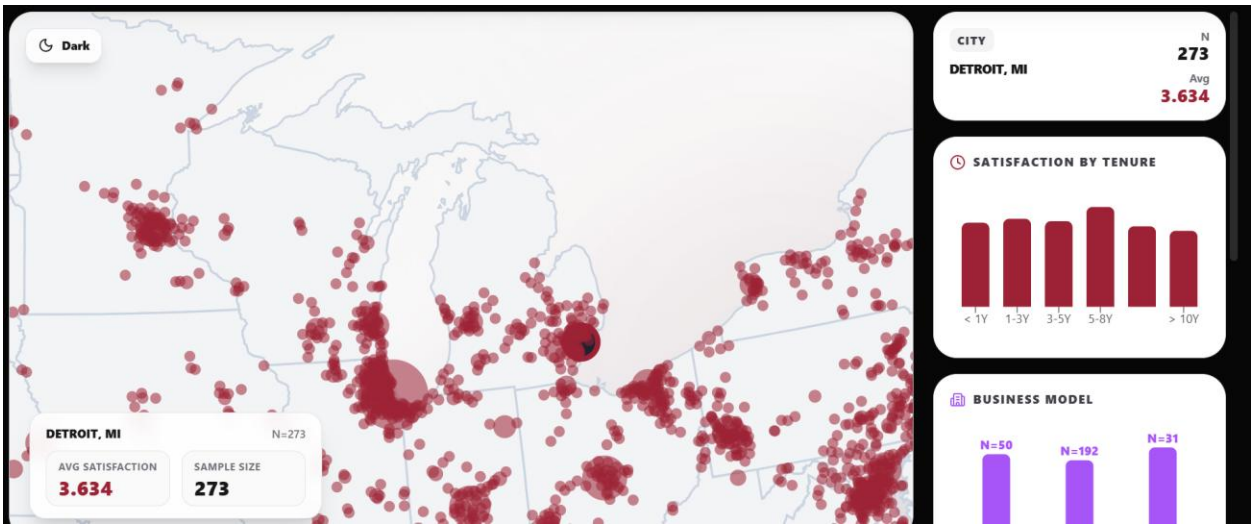
Locations with legislative comparison data display a shield icon.



Option B: Click Directly on the Map

- In City-Level view: click on any circle marker.
- In MSA-Level view: click on any shaded polygon.

The selected location becomes active and updates all charts.



Step 3: Review Summary Metrics

After selecting a location, the top-left summary card displays:

- Location Name
- Satisfaction Score (Average Rating, formatted to three decimals)

• Population (N) – total sample size
• Policy Indicator (shield icon if applicable)

This panel updates automatically with each selection.

Step 4: Legislative Effect Comparison (If Available)

If the selected location has valid policy-period data, the dashboard displays:

• Pre-Law (12–1 months)

• Post-Law (0–11 months)

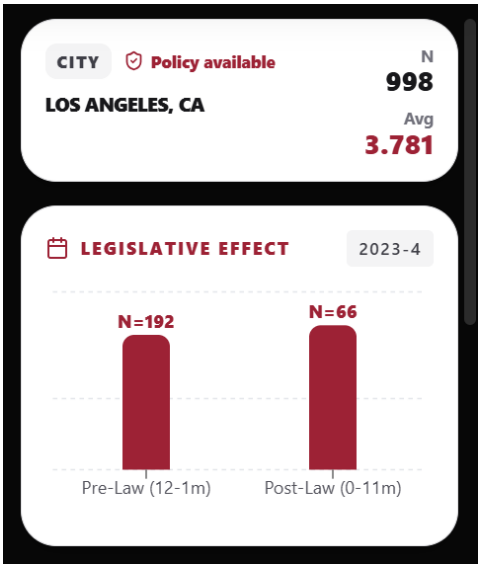
Each bar shows:

• Average satisfaction rating

• Sample size (N) above the bar

If year and month are available, a “Policy Start” label appears.

This section will not display if no valid law comparison data exists.



Step 5: Satisfaction Breakdown by Subgroups

The dashboard provides four subgroup analyses.

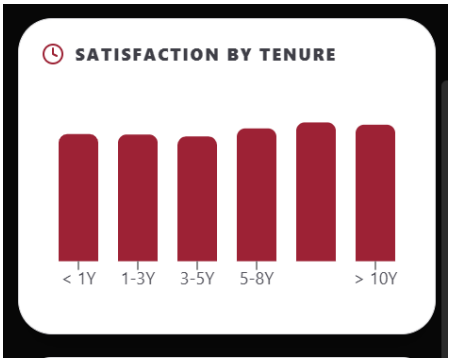
A. Satisfaction by Tenure

Displays satisfaction scores by employee tenure groups:

- < 1Y
- 1–3Y
- 3–5Y
- 5–8Y
- 8–10Y
- > 10Y

Each bar includes tooltip information:

- Average rating
- Sample N



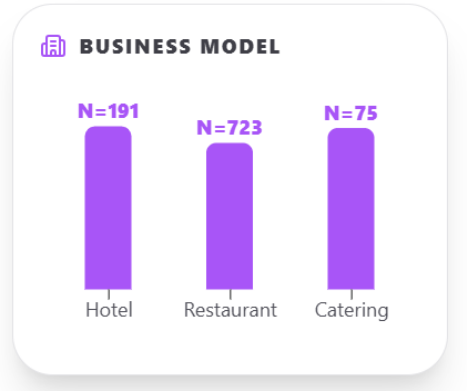
B. Satisfaction by Business Model

Displays:

- Hotel
- Restaurant
- Catering

Each bar includes:

- Average rating
- N value displayed above the bar



C. Satisfaction by Skill Level

Displays:

- Unskilled
- Semi-skilled
- Skilled
- Professional

Each bar includes:

- Average rating
- N value above the bar

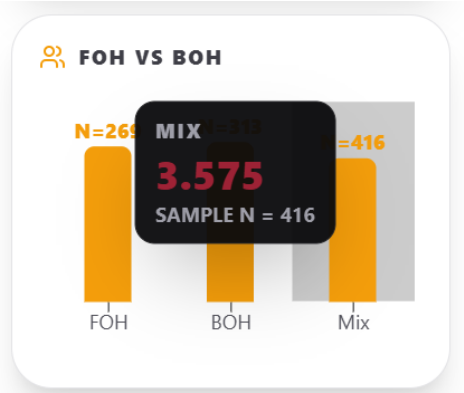
D. Front vs Back-of-House Analysis

Displays:

- FOH (Front-of-House)
- BOH (Back-of-House)
- Mix

Each bar includes:

- Average rating
- N value above the bar



4. Troubleshooting & Error Codes

If the dashboard does not function as expected, check the following:

- “Dashboard Loading…” persists

Cause: One or both CSV files are missing or incorrectly named.

Fix: Verify both required CSV files exist in /public.

- Cities do not appear on the map

Cause: Missing or invalid latitude/longitude fields.

Fix: Ensure latitude and longitude are numeric and non-empty.

• MSA polygons appear but cannot be selected

Cause: GEOID mismatch between msa.json and the MSA CSV.

Fix: Ensure geoid values in CSV match GEOID fields in msa.json.

• Legislative comparison section not visible

Cause: Pre/Post law columns are missing or contain invalid values.

Fix: Confirm valid numeric values exist for law comparison fields.

• State basemap does not load

Cause: Internet connection issue for the external state file.

Fix: Ensure internet access or host the state file locally.